

EGRD: ASSESSING RESEARCH NOVELTY THROUGH EVIDENCE-GROUNDED RESIDUAL DECOMPOSITION

Anonymous authors

Paper under double-blind review

ABSTRACT

Assessing the novelty of a research idea requires identifying what remains beyond prior work, even when coverage is dispersed across several papers or the contribution lies in a new combination of familiar components. We propose Evidence-Grounded Residual Decomposition (EGRD), a supervised framework for predicting ordered novelty grades relative to an explicit set of prior works published before the target. EGRD decomposes the target into latent semantic units and learns a support matrix aligning each unit with each prior. Aggregating support across priors yields unit residuals for insufficiently covered content, while within-prior co-support yields interaction residuals for potentially novel combinations. The model predicts novelty from these residuals together with coverage and contextual representations. To support training and evaluation, we construct IDEADELTA, which pairs research ideas with temporally admissible priors and provides model-generated coverage and novelty annotations across graph machine learning, computer vision, and time series. On temporally held-out ideas, EGRD consistently outperforms prompted baselines across these domains, whether those baselines read the same evidence or select their own priors from the same admissible pool. In a separate transfer experiment, the frozen model evaluated against the expert labels of RINoBench yields higher ordinal agreement and Macro-F1 than the systems that benchmark reports. Ablations show that predictions depend on matched prior evidence and benefit from aggregating support across priors and representing residual contributions.

1 INTRODUCTION

Language models now propose research ideas faster than anyone can vet them, and deciding which of those ideas is new is the step that does not scale. Novelty is not a monotone function of textual similarity to any single prior work. We therefore study *evidence-conditioned* novelty assessment: grading a target idea against an explicit set of prior works published before it, which makes the basis of each judgment inspectable and turns the evidence into a variable we can intervene on.

Most prior work takes a paper as the unit of assessment, from citation-pattern measures (Uzzi et al., 2013) to document-level embeddings (Cohan et al., 2020; Ostendorff et al., 2022) to systems that review a submission as a whole (Zhu et al., 2025). But a paper bundles contributions of uneven novelty, and one score over the bundle does not say which contribution it describes. Systems that do operate on ideas assemble the judgment as a language-model workflow: retrieve related work, then prompt a model to compare against it (Wang et al., 2024; da Silva et al., 2025; Afzal et al., 2026). Here the evidence is selected by the system being evaluated, so no verdict can be attributed to the retrieval or to the comparison, and the comparison itself is elicited by a prompt rather than learned against supervision. Evaluating this arrangement against expert judgments, Schopf & Färber (2026) find that the reasoning such models produce mirrors human rationales, yet that alignment does not translate into accurate grades.

Such a workflow also compresses the comparison into a single score, and three distinctions do not survive. Coverage can be *dispersed*: several papers may jointly account for an idea that no single paper resembles. Ideas at equal coverage differ in *what remains*: one leaves an implementation detail unexplained, another introduces a new mechanism. And individually covered components may form a *combination* that no single prior work puts together. Each distinction asks about a

particular part of the idea and the particular prior work responsible for it, and a score that discards that correspondence cannot be post-processed to recover it. Assessing novelty calls instead for a representation that keeps it.

We propose Evidence-Grounded Residual Decomposition (EGRD), which decomposes a target against ranked historical evidence into what that evidence accounts for and what it leaves, then predicts an ordered novelty grade from the two. Because existing benchmarks do not pair ideas with the evidence admissible when they were proposed, we also construct IDEADELTA to supply that pairing. Our contributions are:

- **IDEADELTA, a multi-domain evidence-paired dataset.** 21,111 ideas paired with temporally admissible priors and annotated at the idea and pair levels (92,876 pairs) across graph machine learning, computer vision, and time series. Supplied rather than retrieved, the evidence can be intervened on: permuting priors while holding targets fixed drops Accuracy from 0.78 to 0.35, the majority-class band.
- **Unit-level coverage over dispersed evidence.** We model coverage at the level of latent semantic units, aligning each unit with each prior through a learned support matrix. Noisy-OR aggregation lets several partially relevant papers jointly cover a unit none covers alone; removing it costs 0.27 quadratic weighted kappa (QWK).
- **Typed residuals.** We split what the priors leave over into unit residuals (content no prior supports) and interaction residuals (combinations no single prior puts together). Both read off the same support matrix, so an unexplored contribution and an effective recombination are distinguished by the model rather than by assertion.
- **Consistent gains across domains.** EGRD reaches 0.909/0.882/0.898 QWK across the three domains against 0.579 for the best prompted system given the same evidence, and 0.292 for the best one that selects its own priors. A separate transfer experiment carries the frozen checkpoint to the expert labels of RINoBench (Schopf & Färber, 2026) in §4.2.

Appendix A situates EGRD against automated idea generation, prior novelty judges, and decomposed verification against evidence.

2 DATASET

Novelty is not a property of an idea alone; it is a property of an idea read against the work that preceded it. Evaluating it therefore requires three things for every example: a target idea, the prior ideas that were available when the target was proposed, and the contribution that survives those priors. Assembling all three together is what IDEADELTA is for.

2.1 TASK AND LABELS

The unit of annotation is a research *idea*: a single innovation claim extracted from a paper. A paper that introduces both a new objective and a new architecture therefore contributes two ideas, annotated separately. For a target idea x published at time t_x , the admissible priors are $\mathcal{P}_x = \{p_k \mid t_{p_k} < t_x\}$. This temporal constraint does real work: an idea published after the target says nothing about the target’s novelty, however similar the two are.

IDEADELTA annotates each target at two levels. At the *pair* level, every target–prior pair receives one of four coverage levels—NOT COVERING, RELATED BUT NOT COVERING, PARTIAL COVER, LARGE COVER—recording how much of the target that single prior accounts for; a target’s retained priors also come with a ranking, which supplies the ordering a model needs when it reads evidence one prior at a time. At the *idea* level, the target receives an ordered novelty grade from LOW to HIGH. The two levels answer different questions: pair labels say what each piece of evidence covers, the idea grade says what survives all of it. A target can hold no strongly covering prior yet still score low, when several partial covers together leave little behind.

The idea-level grade is not an overall impression layered on top of the pair labels. We instead annotate four quantities per target and combine them by the fixed rule $\text{raw}(x) = (1 - C_x) \cdot \Delta_x \cdot E_x \cdot (1 - \lambda S_x)$, $\lambda = 0.45$, where C_x is how much of the target the retained priors jointly cover, Δ_x the substantive contribution that remains, E_x whether the retrieved evidence suffices for a judgment,

and S_x how much of the claimed difference is merely surface-level. Binning $\text{raw}(x)$ yields the four ordered grades. The rule lowers novelty for exactly four reasons, each separately inspectable: prior work already covers the target, the remaining difference is not substantive, the evidence is too thin to judge, or the difference amounts to renaming.

2.2 CONSTRUCTION

Paper selection and idea extraction. We collect arXiv papers in graph machine learning, computer vision, and time series, preferring work that was subsequently published and, among that, work at major venues. An LLM extracts innovation claims from each title, abstract, and introduction. A second pass discards statements that make no substantive research claim—open-source announcements, bare state-of-the-art assertions, and engineering deliverables—so that later stages grade claims rather than boilerplate.

Prior retrieval. For each target, nearest-neighbor search over idea embeddings (Cohan et al., 2020; Karpukhin et al., 2020) returns candidate priors published earlier, and an LLM adjudicates the uncertain similarity band. A cross-type retrieval pass links a target to relevant prior work of a different contribution type, which same-type retrieval alone would record as isolated. The retained priors are ranked for each target, and every pair stays within one research domain.

Annotation. A single judge (Zheng et al., 2023; Li et al., 2025), `deepseek-v4-pro`, annotates all stages. Pair coverage follows a hybrid rule: clearly covering and clearly unrelated pairs take their label directly, while the uncertain band goes to the judge, which reads the two ideas’ technical content. The judge then estimates, for each target, joint coverage, residual contribution, evidence sufficiency, and surface-level difference, and the novelty rule above converts these into the idea-level grade. The labels are model-generated; we inspected samples from every stage. Appendix B.1 gives the full pipeline.

The result is 21,111 annotated ideas and 92,876 target–prior pairs spanning 2019-01 to 2026-09, split by date rather than at random so that no training target postdates an evaluation target; Appendix B.1 reports the composition per domain. Because the labels are model-generated, we also check that they behave as novelty labels should—responding to the retrieved evidence rather than to target metadata alone, and falling when a prior already covers the target contribution—and report those checks in Appendix B.2.

3 METHODOLOGY

Given a target research idea and a set of ideas proposed before it, EGRD assesses the target’s novelty relative to the supplied evidence in three stages (Figure 1). Stage A selects temporally admissible priors and estimates how much of the target they already account for. Stage B decomposes the target against that evidence into two complementary components: unit residuals capture semantic content with insufficient historical support, and interaction residuals capture potentially novel combinations of those units. Stage C predicts an ordered novelty grade from LOW to HIGH from the residual representations. Throughout, $[\cdot; \cdot]$ denotes concatenation, $\odot$ elementwise multiplication, d the encoder dimension, σ the sigmoid, ε a small constant added for numerical stability, and $\mathcal{V}$ the non-padding positions of the target; padded priors are masked out.

3.1 STAGE A: HISTORICAL EVIDENCE INPUTS

Given a target idea x with timestamp t_x , we restrict candidate priors to those predating it, $\mathcal{P}_x = \{p_k \mid t_{p_k} < t_x\}$, and retain the top- K ranked candidates—or all of them, if fewer are available—as the evidence set $\mathcal{E}_x \subseteq \mathcal{P}_x$ ($K = 5$ in our experiments). Each retained prior p_k carries a vector of pair coverage features $\mathbf{f}_k \in \mathbb{R}^{11}$.

To estimate coverage across the evidence set, a cross-encoder (Nogueira & Cho, 2019) encodes each target–prior pair and a multiple-instance model (Ilse et al., 2018) aggregates the pair representations through attention pooling and maximum pooling. Concatenated with evidence-score statistics, the pooled representations predict joint coverage C_J and a four-class coverage distribution π^J . The

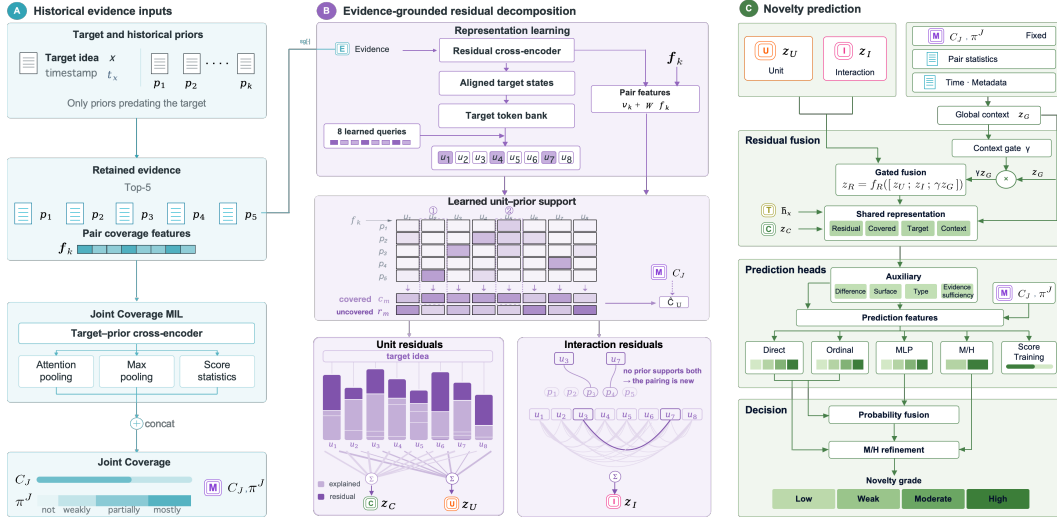


Figure 1: Overview of EGRD. (A) Only priors predating the target are admissible; the retained top- K evidence carries pair coverage features f_k , and a multiple-instance model predicts joint coverage C_J and its four-class distribution π^J . Stage A outputs enter the later stages as fixed inputs, marked $sg[\cdot]$. (B) M learned queries partition a target token bank into semantic units u_m ; a support matrix A_{km} aligns each unit with each prior, yielding the unit residual z_U , the covered pooling z_C , and—through co-support A_{ij} —the interaction residual z_I . (C) A gate modulates global context within residual fusion; the shared representation feeds three novelty branches and the moderate-to-high refinement, while the continuous score head supplies training supervision only.

retained evidence, pair features, and coverage predictions serve as fixed inputs to Stages B and C, grounding the residual decomposition and providing coverage information for novelty prediction.

We summarize the evidence set in a global context vector z_G , a learned projection of twenty-one scalars—the scaled target timestamp, ten pair-score statistics over $\mathcal{E}_x$, and ten joint-coverage features—concatenated with embeddings of two target facets (aspect and contribution type). Apart from the timestamp and the facets, z_G re-encodes the Stage A outputs, and Stage C consumes it as a summary of what the evidence collectively establishes. Appendix C.2 lists the pair and context features and specifies the pair-scoring rule and the joint model.

3.2 STAGE B: EVIDENCE-GROUNDED RESIDUAL DECOMPOSITION

Stage B decomposes the target into semantic units and estimates their support from the retained evidence, representing the unexplained contribution through unit residuals for content with limited historical support and interaction residuals for combinations with limited joint support from any single prior.

Semantic decomposition. A residual cross-encoder, initialized from the Stage A model and trained separately from it, produces a pooled representation v_k for each target–prior pair, which we augment with projected pair coverage features as $\tilde{v}_k = v_k + f_P(f_k)$. Averaging the contextual token states at matching positions across valid pairs yields a shared token bank $\{h_j\}_{j=1}^N$ over the N target tokens. We then use M learned queries q_m to softly assign these tokens to semantic units (Carion et al., 2020; Locatello et al., 2020) ($M = 8$ in our experiments): the queries compete for each token, $B_{mj} = \text{softmax}_m(q_m^\top h_j / \sqrt{d})$, and the resulting assignments are normalized over tokens within each unit, $\alpha_{mj} = B_{mj} / (\sum_{j' \in \mathcal{V}} B_{mj'} + \varepsilon)$, giving $u_m = \sum_{j \in \mathcal{V}} \alpha_{mj} h_j$. The units have no predefined semantic roles. A learned activity score $\phi_m = \sigma(w_\phi^\top u_m + b_\phi)$ controls each unit’s contribution, while an orthogonality penalty on the attention matrix discourages redundant token assignments.

Evidence-aligned unit residuals. A learned support matrix estimates how strongly each prior p_k supports each unit u_m , $A_{km} = \sigma(f_A([u_m; \tilde{v}_k; u_m \odot \tilde{v}_k; |u_m - \tilde{v}_k|; f_k]))$, where f_A is an MLP. Support is estimated separately for every target–prior pair, so different priors can account

for different units. Over the $n = |\mathcal{E}_x|$ retained priors, noisy-OR (Pearl, 1988) aggregates support into unit coverage $c_m = 1 - \prod_{k=1}^n (1 - A_{km})$; its complement, weighted by unit activity, defines the residual weight $r_m = \phi_m(1 - c_m)$ and the unit residual $z_U = M^{-1} \sum_m r_m \mathbf{u}_m$. Noisy-OR lets several partially relevant priors jointly cover a unit that none of them covers alone; we use it as a differentiable aggregation rule, not as a claim that the priors are independent sources of support (Appendix C.3). An active unit with little historical support therefore receives a large residual weight.

Throughout Stage B we preserve the magnitude of the residuals rather than normalize it away. We divide z_U by the fixed number of units M rather than by $\sum_m r_m$, so that the pooled representation retains the total residual mass; the supported content is pooled the same way, as $z_C = M^{-1} \sum_m \phi_m c_m \mathbf{u}_m$. To connect unit coverage to supervision, a learned, activity-weighted pooling of c_m reconstructs joint coverage $\hat{C}_U$, supervised by annotated coverage and the fixed Stage A prediction. The support entries themselves are latent strengths learned without unit-level alignment labels.

Interaction residuals. Units that are individually covered may still form a new combination. For each unit pair $i < j$, an MLP maps their concatenation, elementwise product, and absolute difference to an interaction representation $\mathbf{t}_{ij}$, and a sigmoid head predicts its strength g_{ij} . We measure historical co-support by the largest product of the two units’ support strengths within a single prior, $\bar{A}_{ij} = \max_k A_{ki} A_{kj}$, which gives the interaction residual weight $r_{ij} = g_{ij}(1 - \bar{A}_{ij})$ and the pooled representation $\mathbf{z}_I = \sum_{i < j} \eta_{ij} r_{ij} \mathbf{t}_{ij}$, where the learned attention η_{ij} includes a $\log(r_{ij} + \varepsilon)$ bias that favors pairs with larger residual weights. When different priors support the two units but no single prior strongly supports both, their unit residuals can be small while their interaction residual remains large, provided that g_{ij} is high. Absent co-support is therefore read as an absence of evidence that the combination already exists, which is what a novel combination looks like in the historical record. Co-support only proxies for that evidence: a prior that supports both units need not cover the specific way the target combines them. The pooling keeps the residual magnitude here as well— r_{ij} multiplies $\mathbf{t}_{ij}$ after the attention weights are normalized—and an auxiliary classifier on $\mathbf{z}_I$ is trained with idea-level labels for nontrivial combinations (Appendix C.3).

3.3 STAGE C: NOVELTY PREDICTION

Stage C predicts an ordered novelty grade from the residual representations and the evidence context. Because evidence availability and uncertainty vary across targets, a learned gate γ scales the global context within the residual branch: the gate reads z_U, z_I, z_G and the entropy of the Stage A coverage prediction, and fusion yields $z_R = f_R([z_U; z_I; \gamma z_G])$. Layer normalization over the fused residual, the covered content, the target representation $\mathbf{h}_x$, and the context then gives the shared representation $\mathbf{h} = \text{LayerNorm}([z_R; z_C; \mathbf{h}_x; z_G])$. Since z_G also enters $\mathbf{h}$ directly, the gate controls context only within residual fusion.

Auxiliary heads predict substantive difference, surface change, evidence sufficiency, and residual type. These predictions, together with $\mathbf{h}$ and joint coverage, feed three novelty branches: a direct linear classifier, a cumulative ordinal head that models grade order (Cao et al., 2020), and a two-layer MLP classifier. Each branch produces a four-class distribution, and their convex mixture determines an initial grade.

MODERATE and HIGH account for most of the confusion between grades, so a moderate-to-high refinement revisits that boundary once the initial grade is set. The refinement upgrades an initial MODERATE prediction when a weighted combination of the direct and ordinal HIGH-class probabilities and a separate binary-head score exceeds a threshold. We select the mixture weights and the refinement threshold on development data and fix them for testing (Appendix C.4). A separate continuous novelty head supplies auxiliary supervision during training and stays out of the inference-time grading rule.

Training. We initialize the residual encoder from the Stage A joint-coverage checkpoint and jointly optimize it with the decomposition and prediction modules, while keeping the Stage A models fixed. The objective combines classification of the novelty grade and the residual type, smooth- L_1 supervision of five annotated scalars—substantive difference, surface change, evidence

Table 1: Baselines and our model on the IdeaDelta evaluation set, 3,352 ideas, same labels and the same idea-only protocol (five priors per target, no paper text), split by domain. A row group isolates the effect of the model behind an unchanged prompt chain. The *Dense RAG* group is the one exception to the supplied-evidence protocol: it retrieves its own five priors at query time, and its rows therefore read against the *Direct* rows on the same backbone rather than against the rest of the table. A majority-class predictor scores 0.312, 0.356 and 0.328 Accuracy with QWK zero by construction, and several rows fall below it. Read Accuracy and MAE alongside Macro-F1 and QWK.

System	Backbone	Graph ML				Computer vision				Time series			
		Accuracy	Macro-F1	QWK	MAE ↓	Accuracy	Macro-F1	QWK	MAE ↓	Accuracy	Macro-F1	QWK	MAE ↓
Dense RAG, 5 priors	Qwen3.5	0.360	0.217	0.129	0.867	0.367	0.248	0.191	0.801	0.318	0.248	0.190	0.954
	deepseek-flash	0.371	0.236	0.152	0.842	0.369	0.232	0.154	0.801	0.326	0.224	0.205	0.919
	gpt-5.6-luna	0.362	0.262	0.253	0.810	0.372	0.289	0.267	0.787	0.346	0.295	0.292	0.883
Direct, 5 priors	Qwen3.5	0.300	0.266	0.319	0.925	0.308	0.264	0.300	0.876	0.291	0.283	0.327	0.956
	deepseek-flash	0.438	0.444	0.558	0.706	0.409	0.418	0.521	0.728	0.416	0.415	0.560	0.716
	gpt-5.6-luna	0.432	0.439	0.579	0.657	0.420	0.430	0.561	0.661	0.445	0.436	0.576	0.653
Schopf 2-stage	Qwen3.5	0.324	0.286	0.335	0.831	0.346	0.293	0.338	0.777	0.344	0.333	0.413	0.793
	deepseek-flash	0.443	0.458	0.560	0.639	0.416	0.411	0.501	0.676	0.442	0.447	0.553	0.654
	gpt-5.6-luna	0.391	0.377	0.487	0.698	0.393	0.364	0.458	0.683	0.392	0.379	0.512	0.688
UKPLab 3-stage	Qwen3.5	0.280	0.161	0.125	0.928	0.289	0.150	0.106	0.873	0.274	0.180	0.132	0.973
	deepseek-flash	0.351	0.267	0.301	0.761	0.376	0.272	0.259	0.741	0.386	0.289	0.290	0.727
	gpt-5.6-luna	0.318	0.226	0.266	0.800	0.367	0.248	0.304	0.721	0.365	0.267	0.296	0.743
GraphMind	Qwen3.5	0.264	0.205	0.154	1.027	0.264	0.182	0.108	0.959	0.244	0.204	0.155	1.051
	deepseek-flash	0.291	0.227	-0.007	1.095	0.342	0.266	0.154	0.971	0.365	0.249	0.159	0.916
	gpt-5.6-luna	0.289	0.255	0.046	1.003	0.381	0.328	0.304	0.755	0.380	0.327	0.305	0.750
DeepReviewer	DeepReviewer-7B	0.294	0.237	0.034	0.965	0.325	0.235	-0.043	0.972	0.296	0.222	-0.017	0.964
EGRD (ours)	—	0.799	0.801	0.909	0.201	0.761	0.767	0.882	0.242	0.789	0.776	0.898	0.215

sufficiency, the reconstructed coverage $\hat{C}_U$, and a continuous novelty score—and regularization of the decomposition against redundant token assignments, unused units, and coinciding unit coverages. Because level losses alone leave the scalar heads free to settle on the median of their target, each is also supervised in rank; the coverage term additionally penalizes disagreement with the fixed Stage A prediction, weighted down where that prediction has higher entropy. Appendix C.5 states the full objective, defines each term, and reports the coefficients.

4 EXPERIMENTS

The evaluation set is the IDEADELTA test split. Every system is scored against the same labels by the same script, and all but one receive the same evidence — the five highest-ranked admissible priors per target, as titles and abstracts, with no paper text. The exception is a dense-retrieval baseline that assembles its own five priors at query time, included to measure what the supplied evidence contributes. For the four-level labels with unequal class support, we report Accuracy, Macro-F1, QWK and MAE — exact classification, performance across classes, chance-corrected ordinal agreement and grade error. Appendix D.1 gives the splits, hyperparameters and scoring code. Section 4.2 then takes the frozen checkpoint outside this setting, to expert-labelled ideas under a protocol of their own.

4.1 MAIN RESULTS

We compare against a *direct* single call; two multi-stage prompt chains, *Schopf 2-stage*, the evaluation prompt accompanying RINoBench (Schopf & Färber, 2026), and *UKPLab 3-stage* (Afzal et al., 2026); *GraphMind* (da Silva et al., 2025), which labels each prior as supporting or contrasting before rating; *DeepReviewer-7B* (Zhu et al., 2025), a reviewer model fine-tuned on paper review; and a *dense RAG* group running the *direct* call on five priors it retrieves itself by dense similarity to the target. All but DeepReviewer-7B appear once per backbone. Table 1 reports every system on the IDEADELTA evaluation set.

The benefit of multi-stage prompting depends on the chain and the backbone. Measured against a single five-prior call on the same backbone, the two-stage Schopf chain gains in all three domains on Qwen3.5, by 0.024 to 0.053 Accuracy and 0.016 to 0.086 QWK; on deepseek-flash it stays within 0.026 Accuracy and 0.020 QWK of the direct call in either direction; and on gpt-5.6-luna it loses in all three domains, by 0.027 to 0.053 Accuracy and 0.064 to 0.103 QWK. The three-stage UKPLab chain and GraphMind never reach the direct call at all, sitting below it in all nine cells on both

metrics; the UKPLab QWK deficit widens with the backbone in the same direction, averaging 0.19 on Qwen3.5, 0.26 on deepseek-flash and 0.28 on gpt-5.6-luna. Additional stages therefore buy no consistent advantage: GraphMind spends 476 seconds and 31,201 LLM calls on the 3,352 ideas and still scores below the direct single call.

Which model a chain runs on matters about as much as which chain it is. Over the four chains that share the supplied evidence, holding the chain fixed and changing the model moves Accuracy by 0.103 on average — the mean, across the four chains and three domains, of the range over the three backbones — against 0.091 for the mean range over the four chains at a fixed backbone; on QWK the two swap, 0.195 against 0.348. Neither factor dominates on both metrics, and which of them leads depends on the metric read.

Replacing the supplied priors with query-time dense retrieval costs ordinal agreement on every backbone. Against the direct call on the same backbone and domain, Macro-F1 and QWK each fall in all nine cells, QWK by 0.109 to 0.190 on Qwen3.5, 0.355 to 0.406 on deepseek-flash and 0.284 to 0.326 on gpt-5.6-luna. Accuracy moves the other way on the weakest backbone, rising by 0.027 to 0.060 on Qwen3.5 while Macro-F1 falls, the signature of predictions concentrating on the larger classes: these rows sit at 0.318–0.372 Accuracy, at or just above the majority-class band, with Macro-F1 no higher than 0.295 anywhere. The group also compresses the backbone differences that Table 1 otherwise shows, its Accuracy spanning at most 0.028 in any domain against 0.112 to 0.154 for the direct call. What a system is given to compare against therefore moves the judgment more than which model or chain performs it, which sharpens the evidence check of Appendix B.2: there random priors reduce label prediction to near-chance and establish only that the evidence must be on topic, whereas dense retrieval is on topic and the gap persists, so the candidate lists carry target-prior relationships that generic similarity search does not recover — what the cross-type pass and the adjudicated similarity band (§2) are for.

EGRD outperforms the baselines in all three domains, by 0.356, 0.341 and 0.344 Accuracy and 0.330, 0.321 and 0.322 QWK over the best baseline of each domain, so the advantage is consistent across the three even though individual baselines remain sensitive to the domain. DeepReviewer-7B transfers least: its QWK of 0.034, -0.043 and -0.017 is ordinal agreement close to chance under the supplied-evidence protocol.

4.2 TRANSFER TO EXPERT JUDGMENTS

The comparison above is scored against IDEADELTA’s model-generated labels. To ask what survives outside that setting, we transfer to human expertise. RINoBench (Schopf & Färber, 2026) draws ideas from OpenReview submissions across machine learning as a whole, pairs each with the prior work admissible for it, and has domain experts rate its novelty from 1 to 5; the benchmark then measures how closely language models reproduce those ratings. Its 277 test ideas give a label source independent of ours in both respects that matter here: the grades are human rather than model-generated, and the ideas span the field rather than three of its areas. We evaluate the checkpoint of Table 1 on them frozen, with no retraining and no fine-tuning, folding the expert scale to our four grades by merging 4 and 5 and reading the 25 priors the benchmark supplies per target at the trained input width. The comparison rows are three chains we rerun on the same evidence together with the systems the benchmark itself reports, all rescored on the folded scale by the script that produces Table 1. Appendix D.2 gives the protocol.

EGRD leads both metrics that reward ordering the ideas the way the experts did, and on QWK no other system reaches two thirds of its value while most of the table sits near zero. Accuracy and MAE belong to other rows, but reading those leads as closer agreement mistakes what the folded scale rewards. Merging the top two expert levels leaves 41.5% of the ideas in a single class, so answering with that class alone scores 0.4152 Accuracy and a competitive MAE while ordering nothing, and the systems that lead those two metrics come close to doing exactly that: their ordinal agreement is at or near zero, the value a constant answer earns by construction. Our row reaches the same 0.4152 from a prediction spread over all four grades, 16/36/76/149 against a gold 15/60/87/115, and it is that spread the rank-sensitive metrics register.

The agreement is nonetheless partial, and two properties of the transfer bound it before any model is run. The first is coverage of the literature. IDEADELTA is built from three fields and RINoBench spans machine learning as a whole, so for most of its test ideas the model has never been trained

Table 2: Transfer of the frozen checkpoint to the RINoBench test set, 277 ideas whose expert 1–5 ratings are folded to our four grades. Bold marks the best value in a column.

System	Accuracy	Macro-F1	QWK	MAE ↓
<i>Same Qwen3.5 backbone, zero-shot</i>				
Qwen3.5	0.3574	0.2244	0.1451	0.7726
GraphMind	0.3357	0.2281	0.1469	0.7653
UKPLab 3-stage	0.3466	0.1892	0.0808	0.7220
<i>Released predictions of the benchmark authors, folded to four levels</i>				
GPT-5	0.3646	0.2457	0.0955	0.8051
o3	0.3971	0.2370	0.0606	0.8087
gpt-oss-120b	0.4152	0.2249	0.0392	0.8375
Llama-3.1-8B-Instruct	0.3755	0.1961	−0.0011	0.8773
DeepSeek-R1	0.4224	0.1873	0.0283	0.8700
Llama-4-Scout-17B-16E	0.4116	0.1864	0.0215	0.8809
Llama-3.3-70B-Instruct	0.4188	0.1531	0.0051	0.9025
<i>Reviewer model fine-tuned on paper review, local weights</i>				
DeepReviewer-7B	0.1913	0.1425	0.0145	1.3610
EGRD (ours)	0.4152	0.3407	0.2245	0.8087

Table 3: Ablations on the pooled IDEADELTA evaluation set, 3,352 ideas. **Full** is the unablated checkpoint of Table 1. The pipeline-stage rows intervene on that trained checkpoint at inference and the decomposition-design rows retrain the model without the component, so magnitudes are comparable within a block but not across the two. Higher is better for Accuracy, Macro-F1 and QWK; lower is better for MAE.

System	Accuracy	Macro-F1	QWK	MAE ↓
Full	0.7796	0.7790	0.8958	0.2230
<i>Pipeline stages</i>				
Permuting prior evidence	0.3471	0.3532	0.2241	0.8606
Permuting coverage	0.6649	0.6605	0.5996	0.5702
Zeroing decomposition representations	0.6371	0.6554	0.7878	0.4366
<i>Decomposition design</i>				
Collapsing semantic units	0.6580	0.6292	0.8591	0.4073
Disabling slot competition	0.6701	0.6874	0.7665	0.3985
Removing noisy-OR support	0.6210	0.5520	0.6291	0.6068

to read the work it is now asked to compare against; a prior can only support a unit of a target the model is equipped to parse, and on those ideas the decomposition operates on evidence it cannot ground. The second is the label scale. Transferring a 1–5 expert rating onto four ordered grades requires merging two of its levels, and the merge is not neutral: it erases the distinction the experts drew at the top of their scale, and that is exactly where the mass of the benchmark lies. A model that orders those ideas correctly has no way left to express the difference, and no metric computed on the folded scale can credit it. We therefore read the margin as partial agreement between residual structure learned from model-generated labels and expert judgment, not as the problem solved.

4.3 ABLATION

Because the supplied evidence can be intervened on with the target held fixed, we can test what the trained model’s predictions actually depend on. The three pipeline interventions cut the chain at successive points. **Permuting prior evidence** shuffles the retrieved prior texts across targets, destroying the semantic content the chain is built on; **permuting coverage** shuffles only the joint coverage score and leaves every prior text in place, destroying what the historical evidence inputs (§3.1) infer from that content; **zeroing decomposition representations** sets the unit residual z_U , interaction residual z_I and covered-content representation z_C to zero, removing these three outputs of the decomposition (§3.2) together. All three reduce performance on every metric. Permuting prior evidence costs the most, 0.43 Accuracy and 0.67 QWK, and raises MAE from 0.2230 to 0.8606, dropping Accuracy into the band of the majority-class predictor of Table 1 (0.312–0.356). This drop shows that the trained model relies on evidence matched to the target. Permuting coverage leaves every prior text in place and still costs 0.11 Accuracy, 0.30 QWK and 0.35 MAE, so what the historical evidence inputs infer is not redundant with the evidence it is computed from. Zeroing

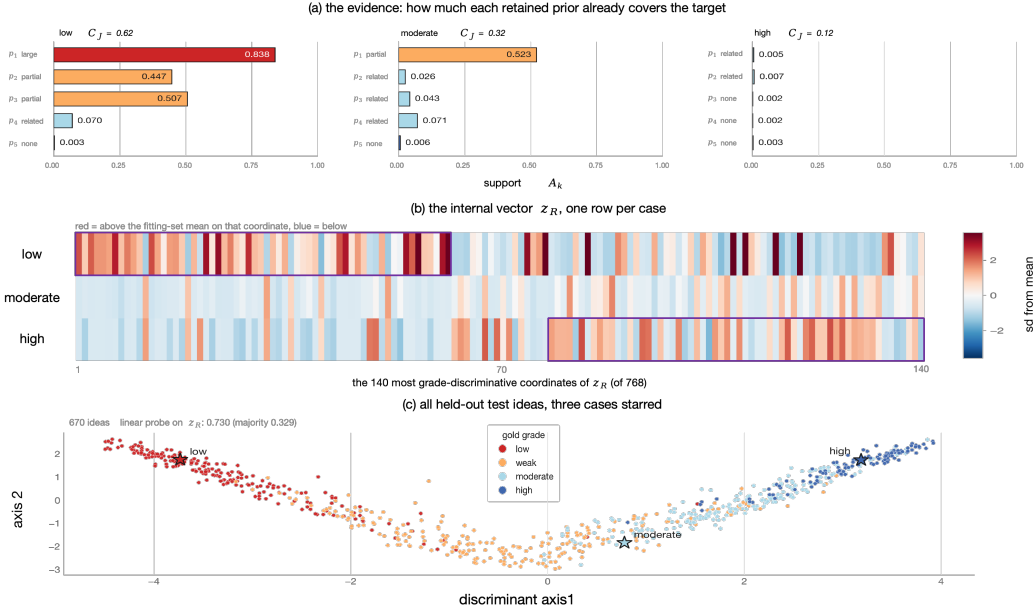


Figure 2: Three test targets, graded low, moderate and high, traced through the model. (a) The support A_k assigned to each retained prior, against the coverage label the annotation carries; C_J is the predicted joint coverage. (b) The residual representation z_R of each case, as deviations from the fitting-set mean. (c) z_R for every held-out test idea, coloured by gold grade, the three cases starred.

the decomposition representations costs 0.14 Accuracy, 0.12 Macro-F1, 0.11 QWK and 0.21 MAE, showing that the trained model benefits from these representations collectively. These last two order differently under the two kinds of metric: zeroing costs more in Accuracy and Macro-F1, permuting coverage more in QWK (0.2962 versus 0.1080) and MAE (0.3472 versus 0.2136), so corrupting the coverage signal damages the grade ordering more even though it misclassifies fewer ideas.

The three decomposition-design ablations leave the pipeline intact and remove one mechanism from the decomposition itself, reducing performance with different effects across metrics. **Collapsing semantic units**, from eight to one, costs 0.12 Accuracy, 0.15 Macro-F1 and 0.18 MAE, while QWK decreases by 0.04 and remains at 0.8591: a single unit still orders the ideas, but the grade it assigns is off by more. **Disabling slot competition**, which removes the softmax through which the units compete for each token, costs 0.11 Accuracy and 0.13 QWK at a comparable MAE (0.3985 against 0.4073), reversing that pattern: deciding which unit a token belongs to is what keeps the units apart in the first place. **Removing noisy-OR support** costs the most of the three on all four metrics, 0.16 Accuracy, 0.23 Macro-F1, 0.27 QWK and 0.38 MAE, supporting the contribution of this aggregation rule to ordinal agreement.

We also sweep the three parameters that govern how much the model is given (the evidence budget K , the training fraction and the encoder input length) one at a time from the default configuration of Table 1, in all three domains. The largest moves Accuracy by at most 0.075, against the 0.34 that separates our row from the best prompted row there, so none of them is what the main comparison turns on; Appendix D.3 reports the sweeps and what they do and do not license.

4.4 CASE STUDY

The ablations show that corrupting the evidence costs Accuracy, but not what the intact pipeline does with it: what the support matrix weights when every retrieved prior is on topic, and whether the residual representation contains linearly accessible grade information. Figure 2 examines both on three correctly predicted test targets. Panel (a) reports the support A_k and the joint coverage C_J the trained model produces for each target, read off the forward pass with nothing fit. Panels (b) and (c) and the probe reported with them hold the three cases out and fit every supervised step on 80% of the remaining test ideas, scoring out of fold (Appendix D.4).

Support follows the annotation rather than the retrieval: the *large cover* prior of the low case takes 0.838, the lone *partial cover* of the moderate case 0.523, and in the high case, whose five priors were all retrieved on the target’s own vocabulary, no support exceeds 0.007. Joint coverage follows at 0.62, 0.32 and 0.12, and so does the mean unit residual, at 0.041, 0.408 and 0.977. The representation z_R contains substantial linearly accessible grade information: a probe using only z_R achieves 0.730 Accuracy against a majority baseline of 0.329, and the held-out ideas lie along the first discriminant axis in grade order. The probe remains below the full model, so the grades are linearly accessible without being linearly separable or the prediction heads redundant.

5 CONCLUSION

EGRD grades an idea by what an explicit set of admissible priors leaves over, estimating coverage at the level of semantic units and predicting the grade from the residuals that survive alignment with the evidence. Supplying that evidence rather than retrieving it is what lets the design be checked: permuting the priors lowers Accuracy to the majority-class band, removing noisy-OR aggregation costs more on every metric than any other change to the decomposition, and a cross-validated probe finds grade information linearly accessible in the residual representation. EGRD outperforms every prompted baseline we evaluate, whether it reads the supplied priors or selects its own. What remains open lies elsewhere (App. E): on expert labels the ordering transfers while the level does not — a gap a reader can locate in the priors themselves.

REFERENCES

- Osama Mohammed Afzal, Preslav Nakov, Tom Hope, and Iryna Gurevych. Beyond “not novel enough”: Enriching scholarly critique with LLM-assisted feedback. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2648–2671, Rabat, Morocco, 2026. Association for Computational Linguistics. URL <https://aclanthology.org/2026.eacl-long.121/>.
- Jinheon Baek, Sujay Kumar Jauhar, Silviu Cucerzan, and Sung Ju Hwang. ResearchAgent: Iterative research idea generation over scientific literature with large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 6709–6738, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.naacl-long.342/>.
- Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, 140:325–331, 2020. doi: 10.1016/j.patrec.2020.11.008.
- Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *Computer Vision – ECCV 2020*, volume 12346 of *Lecture Notes in Computer Science*, pp. 213–229. Springer, 2020. doi: 10.1007/978-3-030-58452-8.13.
- Yuan Chang, Ziyue Li, Hengyuan Zhang, Yuanbo Kong, Yanru Wu, Hayden Kwok-Hay So, Zhi-jiang Guo, Liya Zhu, and Ngai Wong. TreeReview: A dynamic tree of questions framework for deep and efficient LLM-based scientific peer review. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 15651–15682, Suzhou, China, 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.emnlp-main.790/>.
- Arman Cohan, Sergey Feldman, Iz Beltagy, Doug Downey, and Daniel Weld. SPECTER: Document-level representation learning using citation-informed transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 2270–2282, Online, 2020. Association for Computational Linguistics. URL <https://aclanthology.org/2020.acl-main.207/>.
- Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960. doi: 10.1177/001316446002000104.

- Jacob Cohen. Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4):213–220, 1968. doi: 10.1037/h0026256.
- Italo Luis da Silva, Hanqi Yan, Lin Gui, and Yulan He. GraphMind: Interactive novelty assessment system for accelerating scientific discovery. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2025. URL <https://arxiv.org/abs/2510.15706>.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2023. URL <https://arxiv.org/abs/2111.09543>.
- Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 2127–2136, 2018. URL <https://proceedings.mlr.press/v80/ilse18a.html>.
- Yiqiao Jin, Qinlin Zhao, Yiyang Wang, Hao Chen, Kaijie Zhu, Yijia Xiao, and Jindong Wang. AgentReview: Exploring peer review dynamics with LLM agents. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1208–1226, Miami, Florida, USA, 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.emnlp-main.70/>.
- Vladimir Karpukhin, Barlas Öguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6769–6781, Online, 2020. Association for Computational Linguistics. URL <https://aclanthology.org/2020.emnlp-main.550/>.
- Dawei Li, Bohan Jiang, Liangjie Huang, Alimohammad Beigi, Chengshuai Zhao, Zhen Tan, Amrita Bhattacharjee, Yuxuan Jiang, Canyu Chen, Tianhao Wu, Kai Shu, Lu Cheng, and Huan Liu. From generation to judgment: Opportunities and challenges of LLM-as-a-judge. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 2757–2791, Suzhou, China, 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.emnlp-main.138/>.
- Weixin Liang, Yuhui Zhang, Hancheng Cao, Binglu Wang, Daisy Yi Ding, Xinyu Yang, Kailas Vodrahalli, Siyu He, Daniel Scott Smith, Yian Yin, Daniel McFarland, and James Zou. Can large language models provide useful feedback on research papers? a large-scale empirical analysis. *NEJM AI*, 1(8), 2024. doi: 10.1056/AIoa2400196.
- Francesco Locatello, Dirk Weissenborn, Thomas Unterthiner, Aravindh Mahendran, Georg Heigold, Jakob Uszkoreit, Alexey Dosovitskiy, and Thomas Kipf. Object-centric learning with slot attention. In *Advances in Neural Information Processing Systems* 33, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/8511df98c02ab60aealb2356c013bc0f-Abstract.html>.
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024. URL <https://arxiv.org/abs/2408.06292>.
- Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 12076–12100, Singapore, 2023. Association for Computational Linguistics. URL <https://aclanthology.org/2023.emnlp-main.741/>.
- Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019. URL <https://arxiv.org/abs/1901.04085>.

- Malte Ostendorff, Nils Rethmeier, Isabelle Augenstein, Bela Gipp, and Georg Rehm. Neighborhood contrastive learning for scientific document representations with citation embeddings. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 11670–11688, Abu Dhabi, United Arab Emirates, 2022. Association for Computational Linguistics. URL <https://aclanthology.org/2022.emnlp-main.802/>.
- Judea Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, San Mateo, CA, 1988.
- Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zicheng Liu, and Emad Barsoum. Agent laboratory: Using LLM agents as research assistants. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 5977–6043, Suzhou, China, 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.findings-emnlp.320/>.
- Tim Schopf and Michael Färber. Is this idea novel? an automated benchmark for judgment of research ideas. In *Proceedings of the 2026 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC)*, 2026. URL <https://arxiv.org/abs/2603.10303>.
- Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. Can LLMs generate novel research ideas? a large-scale human study with 100+ NLP researchers. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/ea94957d81b1c1caf87ef5319fa6b467-Abstract-Conference.html.
- Brian Uzzi, Satyam Mukherjee, Michael Stringer, and Ben Jones. Atypical combinations and scientific impact. *Science*, 342(6157):468–472, 2013. doi: 10.1126/science.1240474.
- Juraj Vladika and Florian Matthes. Comparing knowledge sources for open-domain scientific claim verification. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2103–2114, St. Julian’s, Malta, 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.eacl-long.128/>.
- David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 7534–7550, Online, 2020. Association for Computational Linguistics. URL <https://aclanthology.org/2020.emnlp-main.609/>.
- David Wadden, Kyle Lo, Lucy Lu Wang, Arman Cohan, Iz Beltagy, and Hannaneh Hajishirzi. MultiVerS: Improving scientific claim verification with weak supervision and full-document context. In *Findings of the Association for Computational Linguistics: NAACL 2022*, pp. 61–76, Seattle, United States, 2022. Association for Computational Linguistics. URL <https://aclanthology.org/2022.findings-naacl.6/>.
- Qingyun Wang, Doug Downey, Heng Ji, and Tom Hope. SciMON: Scientific inspiration machines optimized for novelty. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 279–299, Bangkok, Thailand, 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.acl-long.18/>.
- Miriam Wanner, Seth Ebner, Zhengping Jiang, Mark Dredze, and Benjamin Van Durme. A closer look at claim decomposition. In *Proceedings of the 13th Joint Conference on Lexical and Computational Semantics (*SEM 2024)*, pp. 153–175, Mexico City, Mexico, 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.starsem-1.13/>.
- Jerry Wei, Chengrun Yang, Xinying Song, Yifeng Lu, Nathan Hu, Jie Huang, Dustin Tran, Daiyi Peng, Ruibo Liu, Da Huang, Cosmo Du, and Quoc V. Le. Long-form factuality in large language models. In *Advances in Neural Information Processing Systems 37*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/937ae0e83eb08d2cb8627fe1def8c751-Abstract-Conference.html.

- Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. C-Pack: Packed resources for general Chinese embeddings. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 641–649, 2024. doi: 10.1145/3626772.3657878.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track*, 2023. URL <https://arxiv.org/abs/2306.05685>.
- Tianshi Zheng, Zheyang Deng, Hong Ting Tsang, Weiqi Wang, Jiaxin Bai, Zihao Wang, and Yangqiu Song. From automation to autonomy: A survey on large language models in scientific discovery. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 17733–17750, Suzhou, China, 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.emnlp-main.895/>.
- Minjun Zhu, Yixuan Weng, Linyi Yang, and Yue Zhang. DeepReview: Improving LLM-based paper review with human-like deep thinking process. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 29330–29355, Vienna, Austria, 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.acl-long.1420/>.

A RELATED WORK

Automated research and idea generation. Language models are increasingly used to carry out parts of the research process itself. Wang et al. (2024) retrieve inspirations from past literature and optimise explicitly for novelty by comparing a candidate against prior papers; Baek et al. (2025) refine proposals against a set of LLM reviewing agents that give structured feedback along several dimensions; and Lu et al. (2024) and Schmidgall et al. (2025) close the loop from ideation through experiments to a written paper and a simulated review. Si et al. (2025) evaluate what such pipelines produce, finding the generated ideas rated more novel than human-written ones but less feasible; Zheng et al. (2025) survey the progression of these systems toward greater autonomy. Every one of these systems contains an internal novelty check, and it is the check rather than the generator that bounds them: once proposals are cheap, the output of the pipeline is whatever its filter admits. The filter is nonetheless the component these systems evaluate least directly, judged only through the ideas that eventually emerge downstream of it.

Judging the novelty of scientific work. Studied on its own, the check has passed through several generations of method. The earliest operate on citation statistics, measuring how unusual the combinations of references in a paper are against historical co-occurrence (Uzzi et al., 2013); they require the work to already exist and be cited, and reduce novelty to patterns of co-occurrence rather than to what a paper says. Document-level semantic representations replace surface statistics with learned relatedness, scoring how close a submission is to prior papers through citation-informed embeddings (Cohan et al., 2020; Ostendorff et al., 2022), but they compare documents as points and recover nothing about which content is shared. Most recent systems retrieve related literature and hand the comparison to a language model. Liang et al. (2024) obtain structured feedback from a single prompt, of which significance and novelty is one section, and report partial overlap with human reviews; Zhu et al. (2025) train a reviewer model on a multi-stage process of structured analysis, retrieval, and evidence-based argumentation; Chang et al. (2025) decompose the review into a tree of questions resolved from leaf to root, and Jin et al. (2024) simulate the review process itself to expose how much of the outcome is attributable to reviewer bias rather than to the paper; Afzal et al. (2026) split the judgment into claim extraction, retrieval and synthesis of related work, and structured comparison, motivating the design with an analysis of human-written novelty reviews; and da Silva et al. (2025) build an interactive system on the same retrieve-then-compare pattern. Schopf & Färber (2026) supply the evaluation counterpart and report that model ratings still diverge substantially from expert ones even when the accompanying reasoning appears sound. Two properties persist across these generations, and both are structural rather than incidental. The evidence is retrieved by the system under evaluation, so its verdicts are unattributable: an error cannot be

assigned to the retrieval or to the judgment, and the two therefore cannot be improved, or even measured, separately. We remove this by conditioning on a given, temporally admissible evidence set, which additionally turns the evidence into a variable that can be intervened on. The second property is that the target is scored as an undivided whole, against the closest prior work or against a retrieved set read as a block. Such a comparison returns one number per target, and no post-processing of that number can recover which part of the idea it is about; the information is discarded before the score is formed, not lost afterwards.

Decomposed verification against evidence. Saying which part remains requires decomposing the target, and outside novelty assessment this has become the standard recipe for factuality evaluation: break a text into smaller propositions and check each against retrieved evidence. Min et al. (2023) decompose a passage into atomic facts, verify each against a knowledge source, and report the supported fraction; Wei et al. (2024) scale the same procedure with search as the evidence source; and in the scientific domain the corresponding task checks a stated claim against candidate abstracts while identifying the supporting rationales (Wadden et al., 2020; 2022), where the choice of evidence source itself moves the verdict (Vladika & Matthes, 2024). Three properties of the recipe do not survive the move to research ideas. The propositions are assumed to be given, whereas an idea carries no stated claims, and the decomposition step is not a neutral preprocessor: Wanner et al. (2024) show that scores of this family shift materially with the decomposition used. Support is treated as a label, whereas a prior work can cover part of a unit and not the rest. And each unit is verified independently before the results are averaged, so the score is a function of the per-unit coverages alone and cannot express whether a combination of units has precedent. We accordingly induce the units rather than reading them off the text, measure support as a graded quantity, and add a residual defined over pairs of units.

B DATASET DETAILS

B.1 DATASET CONSTRUCTION PIPELINE

Sources and idea extraction. Papers are drawn from arXiv in three areas—graph machine learning, computer vision, and time series. Selection prefers work that was subsequently published and, among that, papers at major venues. From each paper’s title, abstract and introduction an LLM extracts innovation claims and assigns each an innovation aspect; a second LLM pass discards claims that are not innovation claims at all—open-source and reproducibility statements, bare state-of-the-art assertions, and engineering deliverables—so that later stages judge substantive claims rather than boilerplate.

Candidate priors. Idea texts are embedded and indexed for top- K nearest-neighbour recall, and candidate relations are formed under a hybrid rule: high similarity is accepted directly, low similarity is rejected directly, and the uncertain band is passed to an LLM. The base build pairs ideas of the same type only, which would record an idea whose closest relative is of another type—a theoretical contribution nearest to a method—as isolated; a cross-type pass recovers these. The retained priors are ranked for each target. Pairing stays within a source domain, and no cross-domain pair is labelled, so the three single-domain subsets and their union address the same ideas under identifiers namespaced by domain. The assembled dataset passes its own validation checks with no duplicate pair keys, no duplicate novelty targets, no invalid pair labels, no target left without a pair, and no temporal violation.

Annotation. All annotation stages use the same judge, `deepseek-v4-pro`. It labels pair coverage over the four classes, with the ambiguous band—related but not covering, partial cover, large cover, and high-similarity pairs the graph left unconnected—re-judged rather than accepted from the similarity rule. It then estimates, for each target, the substantive residual that survives the retained evidence, the sufficiency of that evidence, and the extent to which the claimed difference is surface-level. Idea-level grades follow by applying the novelty rule of Sec. 2.1 to those quantities. Labels are model-generated; each stage was reviewed by sampled manual inspection.

Table 4: Composition of IDEADELTA: counts of ideas and of target–prior pairs, and the idea-level novelty distribution, by source domain. Pairs are formed within a source domain only, so *All* is the union of the three subsets rather than a fourth collection.

	Graph ML	Computer vision	Time series	All
Ideas	5,225	8,291	7,595	21,111
Pairs	21,183	36,238	35,455	92,876
LOW	819	901	1,724	3,444
WEAK	1,240	1,711	1,614	4,565
MODERATE	2,247	2,777	2,173	7,197
HIGH	919	2,902	2,084	5,905

Composition. IDEADELTA contains 21,111 annotated ideas and 92,876 annotated target–prior pairs, with target dates from 2019-01 to 2026-09 (Table 4). Each idea carries its source domain, and the *All* column is the union of the three domain subsets rather than a fourth collection.

Splits. Splits are date-ordered rather than random, so that no training target is dated after an evaluation target. The cuts fall at 2025-02 and 2025-11, giving 13,815/3,944/3,352 ideas for training, validation, and test.

B.2 VALIDATION OF RESEARCH NOVELTY LABELS

Our novelty labels assess the substantive contribution of a target idea beyond its retrieved prior work. We validate this interpretation through complementary experiments. First, we examine how well metadata and retrieved prior content predict the existing labels. We then modify the evidence and repeat annotation to test whether novelty decreases when the target contribution is already covered and remains stable when its substantive contribution is unchanged.

Prediction from metadata. We first test whether simple metadata can account for the novelty labels without describing the technical content of prior work. We consider six variables: character length, word count, publication year, the innovation aspect and contribution type, and the number of retrieved priors. Over all 21,111 ideas, we measure their individual associations with the novelty grade using Spearman ρ for continuous variables and ε^2 for categorical variables. We set the thresholds in advance at $|\rho| \geq 0.20$ and $\varepsilon^2 \geq 0.06$.

Text length, word count, and publication year show weak associations ($\rho = +0.049$, $+0.058$, and -0.130 , respectively), while the innovation aspect and contribution type have $\varepsilon^2 \leq 0.033$. All remain below the corresponding thresholds. The number of priors and their mean coverage are negatively correlated with novelty ($\rho = -0.286$ and -0.471 , respectively). The negative association with coverage is consistent with the intended meaning of the labels: contributions already covered by prior work receive lower novelty grades.

Individual associations may overlook information available from several variables together. We therefore train a classifier on all six metadata variables to predict the four novelty grades. We use the date-ordered split of Appendix B.1, with 13,815 training ideas and 3,352 test ideas, and evaluate quadratic weighted kappa (QWK), which accounts for the ordering of the grades. For comparison, we train the same classifier using only mean prior coverage. QWK increases from 0.227 with metadata to 0.539 with coverage alone. Coverage thus predicts the labels more accurately than the metadata considered here, supporting the relevance of the relationship between the target and prior work.

Prediction with retrieved and random priors. We next examine whether the specific retrieved content matters for label prediction. In the TRUE condition, each target is paired with its five retrieved priors. In the RANDOM condition, these are replaced by five ideas sampled uniformly from those published before the target. The two conditions use the same number of priors and total evidence text volume, and all priors precede the target. We hold the model, hyperparameters, date-ordered split, and random seeds fixed. Models are trained separately from scratch and evaluated on the same held-out targets against their original labels.

Across the three seeds, QWK decreases from 0.196 with retrieved priors to 0.005 ± 0.021 with random priors. The gap of 0.192 is about an order of magnitude larger than the between-seed

standard deviation of 0.015. Replacing retrieved priors with random earlier work therefore reduces prediction to near-chance agreement. This comparison supports the importance of the retrieved content, beyond the amount of evidence supplied.

Response to evidence covering the target contribution. The prediction experiments keep the labels fixed. We now test whether the annotation procedure responds appropriately when the evidence changes. If an earlier study already contains the target contribution, the assigned novelty should decrease. We sample 30 targets across the four grades and construct one synthetic prior for each target. This prior describes the same technical contribution and is dated before the target. After adding it to the evidence set, we repeat the complete annotation procedure: pair coverage labeling, residual contribution estimation, and grading according to the novelty rule of Sec. 2.1. Synthetic priors have dedicated identifiers and are marked `synthetic_control`. They do not refer to real papers, authors, or DOIs and are excluded from the released dataset.

Each modified target is compared with five independent annotations of its original evidence set to account for variation across repeated annotations. We evaluate recognition of the added prior as large coverage, changes in joint coverage and residual contribution, and decreases in the novelty score and grade. Ties are excluded from the exact sign test.

The added prior is labeled large coverage in all 29 comparable cases. Joint coverage increases, while residual contribution and novelty score decrease. The novelty grade decreases for 82.8% of targets (sign test $p = 2.2 \times 10^{-7}$). These results show that the annotation procedure lowers novelty when the evidence covers the target contribution.

Stability under changes that preserve the contribution. A novelty judgment should also remain stable when the evidence changes without altering the substantive contribution of the target. For the same 30 targets, we reorder the priors, duplicate existing priors, or add unrelated papers. The novelty grades before and after these changes show high agreement: Cohen’s κ (Cohen, 1960) = 0.970, 0.902, and 1.000, respectively. These results complement the coverage experiment: the judgments respond to evidence that reduces the remaining contribution and remain comparatively stable under changes that preserve it.

C METHOD DETAILS

C.1 NOTATION AND DIMENSIONS

Table 5 collects the symbols of all three stages. Every vector representation lies in $\mathbb{R}^d$, so the MLPs take inputs of dimension $4d+11$ for f_A , $4d$ for f_I and $3d$ for f_R , and the shared representation $\mathbf{h}$ has dimension $4d$. Section 3 defines the symbols themselves; the remainder of this appendix supplies the components each stage leaves unspecified.

C.2 HISTORICAL COVERAGE MODELS

Pair coverage and evidence selection. For each target–prior pair (x, p_k) , the pair coverage model estimates how much of the target the prior accounts for. A cross-encoder predicts relatedness a_k , coverage conditional on relatedness b_k , and large coverage conditional on coverage d_k , which factorize into a distribution over four ordered classes—unrelated, related but uncovered, partially covered, and largely covered:

$$\pi_k^{\text{pair}} = [1 - a_k, a_k(1 - b_k), a_k b_k(1 - d_k), a_k b_k d_k]. \quad (1)$$

Indexing the classes from 0 to 3, the model scores the pair as $s_k = \frac{1}{2}\pi_{k,2}^{\text{pair}} + \pi_{k,3}^{\text{pair}}$, counting partial coverage at half the weight of large coverage. The pipeline ranks temporally admissible candidates by $s_k + 0.02\ell_k/3$, where ℓ_k is the predicted ordinal coverage label, and retains at most K of them. The retained priors and their pair coverage features form the evidence inputs to the subsequent stages.

Pair feature vector. The features carried into Stages B and C are

$$\mathbf{f}_k = [s_k; s_k + 0.02\ell_k/3; \pi_k^{\text{pair}}; a_k; a_k b_k; d_k; r_k/K; \mathbb{1}[\text{prior text available}]] \in \mathbb{R}^{11}, \quad (2)$$

Table 5: Symbols used in Stages A–C. Indices: k ranges over retained priors, m over semantic units, j over target tokens, and $i < j$ over unit pairs.

Symbol	Meaning	Size
<i>Sizes and index sets</i>		
d	encoder dimension	768
K	retained priors per target	5
M	semantic units	8
N	target tokens	—
n	retained priors, $ \mathcal{E}_x $	$\leq K$
$\mathcal{E}_x$	evidence set of target x	—
$\mathcal{V}$	non-padding target positions	—
<i>Stage A: historical evidence</i>		
π_k^{pair}	pair coverage distribution	$\mathbb{R}^4$
s_k	pair coverage score	$[0, 1]$
$\mathbf{f}_k$	pair coverage features	$\mathbb{R}^{11}$
$\mathbf{h}_k^J$	pair representation (joint model)	$\mathbb{R}^d$
$\mathbf{z}_J$	bag representation	$\mathbb{R}^d$
π^J	joint coverage distribution	$\mathbb{R}^4$
C_J	scalar joint coverage	$[0, 1]$
$\mathbf{z}_G$	global context	$\mathbb{R}^d$
<i>Stage B: residual decomposition</i>		
$\mathbf{v}_k, \tilde{\mathbf{v}}_k$	pair representation, augmented	$\mathbb{R}^d$
$\mathbf{h}_j$	target token state	$\mathbb{R}^d$
$\mathbf{q}_m$	learned unit query	$\mathbb{R}^d$
$\mathbf{u}_m$	semantic unit	$\mathbb{R}^d$
ϕ_m	unit activity	$[0, 1]$
A_{km}	support of prior k for unit m	$[0, 1]$
c_m	unit coverage (noisy-OR)	$[0, 1]$
r_m	unit residual weight	$[0, 1]$
$\hat{C}_U$	reconstructed joint coverage	$[0, 1]$
$\mathbf{t}_{ij}$	interaction representation	$\mathbb{R}^d$
g_{ij}	interaction strength	$[0, 1]$
A_{ij}	co-support, $\max_k A_{ki} A_{kj}$	$[0, 1]$
r_{ij}	interaction residual weight	$[0, 1]$
η_{ij}	interaction attention	$[0, 1]$
$\mathbf{z}_U, \mathbf{z}_I, \mathbf{z}_C$	unit, interaction, covered pooling	$\mathbb{R}^d$
<i>Stage C: novelty prediction</i>		
$\mathbf{h}_x$	target representation	$\mathbb{R}^d$
γ	context gate	$[0, 1]$
$\mathbf{z}_R$	fused residual	$\mathbb{R}^d$
$\mathbf{h}$	shared representation	$\mathbb{R}^{4d}$

where r_k is the rank of the prior within $\mathcal{E}_x$. Through Eq. (1), the first seven entries are deterministic functions of (a_k, b_k, d_k) , so $\mathbf{f}_k$ carries three degrees of freedom from the pair model alongside the two ranking quantities and the availability flag. We keep the redundant parameterization because the downstream projections consume these quantities at different scales.

Joint coverage. Different priors may cover different parts of the target, so a single pair score understates what the evidence establishes together. We therefore estimate joint coverage with a multiple-instance model that treats each pair (x, p_k) as an instance and the retained evidence set as a bag (Ilse et al., 2018). Let $n = |\mathcal{E}_x|$ denote the number of retained priors. A shared cross-encoder maps each pair to a representation $\mathbf{h}_k^J$, from which the model computes attention weights β_k and internal evidence scores e_k :

$$\beta_k = \text{softmax}_k(\mathbf{w}_a^\top \mathbf{h}_k^J + b_a), \quad e_k = \sigma(\mathbf{w}_e^\top \mathbf{h}_k^J + b_e). \quad (3)$$

For a nonempty evidence set, noisy-OR pools the evidence scores as $e_\cup = 1 - \prod_{k=1}^n (1 - e_k)$. Four statistics summarize the scores—that aggregate, their maximum and mean, and the fraction of the K evidence slots that are filled—and join the attention-pooled and elementwise maximum-pooled

pair representations:

$$\mathbf{v}_E = \left[e_{\cup}, \max_k e_k, \frac{1}{n} \sum_{k=1}^n e_k, \frac{n}{K} \right], \quad (4)$$

$$\mathbf{z}_J = f_J \left(\left[\sum_{k=1}^n \beta_k \mathbf{h}_k^J; \max_k \mathbf{h}_k^J; \mathbf{v}_E \right] \right), \quad (5)$$

where f_J is an MLP. Attention pooling aggregates evidence across priors, while maximum pooling retains the strongest feature values; the noisy-OR aggregate enters only as one of the four statistics, and the prediction itself is learned from the full bag representation. A hierarchical head maps $\mathbf{z}_J$ to three probabilities: any coverage, substantial coverage conditional on any coverage, and mostly covered conditional on substantial coverage. Under the same factorization as Eq. (1), these probabilities yield π^J over four joint-coverage classes mirroring the pair classes: *not*, *weakly*, *partially*, and *mostly* covered. The scalar estimate is $C_J = \sum_{\ell=0}^3 \mu_{\ell} \pi_{\ell}^J$, where μ_{ℓ} is the mean annotated coverage score for class ℓ on the training split. Both C_J and π^J are supplied as fixed inputs to the residual and novelty modules.

Global context features. The twenty-one scalars projected into $\mathbf{z}_G$ are the scaled target timestamp, ten pair-score statistics, and ten joint-coverage features. The pair-score statistics over $\mathcal{E}_x$ are the fraction of the K evidence slots that are filled; the maximum, mean, standard deviation and sum of the s_k ; a union score over them; the fractions of retained priors judged covering and judged related; the margin between the top two ranking scores; and the entropy of the pair scores. The joint-coverage features are two scalar coverage estimates, the raw and normalized entropies of π^J , an indicator for whether a joint prediction is available, the gap between the joint estimate and the pair-score union, and π^J itself.

C.3 RESIDUAL REPRESENTATIONS AND POOLING

This section specifies the encoders, pooling rules, and masking behavior that Section 3.2 leaves implicit.

Pair encoding. The residual cross-encoder shares parameters across all retained priors. For each target–prior pair (x, p_k) it produces token states $\mathbf{H}_k \in \mathbb{R}^{L \times d}$, where L is the input sequence length. Masked mean pooling over these states gives the pair representation, which we augment with projected pair coverage features:

$$\mathbf{v}_k = \text{MeanPool}(\mathbf{H}_k), \quad \tilde{\mathbf{v}}_k = \mathbf{v}_k + f_P(\mathbf{f}_k), \quad (6)$$

where f_P consists of a linear map, layer normalization, GELU, and dropout. The target token bank averages target-token states at matching positions across valid pairs, and masked mean pooling over the bank yields the target representation $\tilde{\mathbf{h}}_x$ that Stage C fuses.

Coverage reconstruction. To supervise the unit coverage estimates, we aggregate them into a scalar joint-coverage prediction. The pooling weights depend on both the semantic representation and the activity of each unit:

$$\hat{C}_U = \sum_{m=1}^M \omega_m c_m, \quad \omega_m = \frac{\phi_m \exp(\mathbf{w}_c^\top \mathbf{u}_m + b_c)}{\sum_{r=1}^M \phi_r \exp(\mathbf{w}_c^\top \mathbf{u}_r + b_c)}. \quad (7)$$

Annotated coverage and the fixed Stage A prediction supervise the resulting estimate $\hat{C}_U$. This supervision is aggregate: it constrains the unit coverages without determining the latent support matrix uniquely. Noisy-OR likewise serves as a differentiable rule for aggregating support, and using it does not establish that support from different priors is statistically independent.

Interaction encoding and attention. For each unit pair $i < j$, an MLP f_I encodes the two units and their elementwise comparisons into an interaction representation. A sigmoid head then predicts the interaction strength:

$$\mathbf{t}_{ij} = f_I([\mathbf{u}_i; \mathbf{u}_j; \mathbf{u}_i \odot \mathbf{u}_j; |\mathbf{u}_i - \mathbf{u}_j|]), \quad g_{ij} = \sigma(\mathbf{w}_g^\top \mathbf{t}_{ij} + b_g). \quad (8)$$

Using the interaction residual weights r_{ij} defined in Stage B, we compute attention over all $\binom{M}{2}$ unit pairs:

$$\eta_{ij} = \text{softmax}_{i < j} (\mathbf{w}_I^\top \mathbf{t}_{ij} + b_I + \log(r_{ij} + \varepsilon)), \quad \mathbf{z}_I = \sum_{i < j} \eta_{ij} r_{ij} \mathbf{t}_{ij}. \quad (9)$$

The logarithmic bias directs attention toward pairs with larger residual weights. Multiplying by r_{ij} after normalization then restores their scale, so that weak residuals contribute less than the softmax alone would allow—the same reason $\mathbf{z}_U$ is normalized by M rather than by $\sum_m r_m$.

Masking and edge cases. Padded evidence slots carry zero support: the evidence-availability flag masks the support matrix, which is clamped to $[0, 0.995]$ before use. Both aggregations remain well defined when fewer than K priors are retained. The noisy-OR product runs over the retained priors only, while the co-support maximum is taken over all K slots, so $\bar{A}_{ij} = 0$ when none is retained—an empty record yields no evidence that the combination exists.

C.4 RESIDUAL FUSION, PREDICTION HEADS, AND DECISION RULES

This section specifies the gate, the heads, and the decision rules that Section 3.3 states only in outline.

Context gate. The fusion gate conditions the contribution of global context on the two residual representations and the uncertainty of the joint-coverage prediction. Let H_J denote the normalized entropy of π^J . The gate is

$$\gamma = \sigma(f_\gamma([z_U; z_I; z_G; H_J])). \quad (10)$$

Here f_γ is a two-layer MLP with a $d/2$ bottleneck and a scalar output, so γ scales $\mathbf{z}_G$ uniformly rather than gating it per dimension. The gated context $\gamma \mathbf{z}_G$ enters residual fusion alongside $\mathbf{z}_U$ and $\mathbf{z}_I$, while $\mathbf{z}_G$ reaches the shared representation—and through it the prediction heads—on an ungated path of its own.

Auxiliary predictions. The auxiliary heads use the shared representation $\mathbf{h}$ to predict substantive difference $\hat{d}$, surface change $\hat{s}$, evidence sufficiency $\hat{e}$, and residual-type probabilities π^T . The first three outputs are sigmoid scores; the type prediction uses the hierarchy described below.

Residual types. The type hierarchy distinguishes a minor variant from six substantive families: conceptual innovation, mechanism or system, task or application, training or evaluation, data-related innovation, and combination. Conditional fine heads yield eleven types in total. The minor/substantive gate additionally reads $\hat{d}$, $\hat{s}$, and $\max_{i < j} r_{ij}$. The probability assigned to the combination type is multiplied by $0.70 + 0.30\sigma(f_{\text{comb}}(\mathbf{z}_I))$ before the hierarchical distribution is renormalized. The final type distribution is a convex mixture of the hierarchical and flat predictions, with flat weight 0.15.

Novelty branches. All novelty branches receive the prediction features

$$\xi = [\mathbf{h}; \hat{d}; \hat{s}; \hat{e}; \pi^T; C_J; \pi^J]. \quad (11)$$

We encode LOW, WEAK, MODERATE, and HIGH as 0, 1, 2, and 3, respectively. A direct linear classifier and a two-layer MLP classifier map ξ to class distributions $\mathbf{p}^D$ and $\mathbf{p}^R$.

The ordinal branch models cumulative probabilities of reaching each grade threshold. It produces three sigmoid gates g_1, g_2, g_3 from ξ and defines

$$q_1 = g_1, \quad q_2 = g_1 g_2, \quad q_3 = g_1 g_2 g_3, \quad \mathbf{p}^O = [1 - q_1, q_1 - q_2, q_2 - q_3, q_3]. \quad (12)$$

Here q_ℓ estimates the probability that $y \geq \ell$, with $q_1 \geq q_2 \geq q_3$ by construction.

Probability fusion. We aggregate the three branch distributions through a convex mixture:

$$\mathbf{p} = \lambda_D \mathbf{p}^D + \lambda_O \mathbf{p}^O + \lambda_R \mathbf{p}^R, \quad \lambda_D + \lambda_O + \lambda_R = 1, \quad \lambda_D, \lambda_O, \lambda_R \geq 0. \quad (13)$$

The fused distribution provides the initial grade for the refinement below.

Moderate-to-high refinement. A separate binary head $h_{MH} = \sigma(f_{MH}(\xi))$ scores the distinction between moderate and high novelty. At inference, let $y_0 = \arg \max_{\ell} p_{\ell}$ for the mixture in Eq. (13), and define

$$h_{\text{high}} = \alpha_D p_3^D + \alpha_O p_3^O + \alpha_{MH} h_{MH}, \quad \hat{y} = \begin{cases} 3, & y_0 = 2 \text{ and } h_{\text{high}} \geq \theta_{MH}, \\ y_0, & \text{otherwise.} \end{cases} \quad (14)$$

where $\alpha_D, \alpha_O, \alpha_{MH} \geq 0$ and $\alpha_D + \alpha_O + \alpha_{MH} = 1$. Thresholding the weighted score, rather than requiring separate agreement from all three heads, promotes a MODERATE prediction to HIGH whenever the heads are collectively confident—even when each of them favors a different grade. We select the mixture weights and the refinement threshold on development data and fix them for testing; Appendix C.5 reports the values.

Continuous outputs. A separate sigmoid head predicts a continuous novelty score. Probability fusion and the moderate-to-high refinement alone determine the grade at inference, so this score never enters the decision rule.

C.5 TRAINING AND IMPLEMENTATION DETAILS

This section defines the loss terms of the training objective and reports the settings of the residual stage.

Objective. Let $\mathcal{A} = \{\Delta, S, E, C, \nu\}$ index the annotated scalars—the substantive difference, surface change, and evidence sufficiency read off the auxiliary heads, the reconstructed coverage $\hat{C}_U$, and the continuous novelty score. Each contributes a smooth- L_1 level loss $\mathcal{L}_a$ against its annotation, and the ranking terms of Δ, S, E and C are collected into a single $\mathcal{L}_{\text{rank}}$, the novelty score carrying its own inside $\mathcal{L}_{\nu}$:

$$\mathcal{L} = \sum_{a \in \mathcal{A}} \lambda_a \mathcal{L}_a + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{nov}} \mathcal{L}_{\text{nov}} + \lambda_{\text{type}} \mathcal{L}_{\text{type}} + \lambda_{\text{dec}} \mathcal{L}_{\text{dec}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}}, \quad (15)$$

Here $\mathcal{L}_{\text{nov}}$ supervises the three novelty branches, their fused distribution, and the moderate-to-high head; $\mathcal{L}_{\text{type}}$ supervises residual categories and nontrivial combinations; and $\mathcal{L}_{\text{dec}}$ and $\mathcal{L}_{\text{var}}$ regularize the decomposition. Each coefficient λ weights its term in the sum.

Supervision. $\mathcal{L}_{\text{nov}}$ trains the direct and MLP classifiers with class-weighted cross-entropy, the ordinal outputs against $\mathbf{1}[y \geq \ell]$, $\ell = 1, 2, 3$, with weighted binary cross-entropy, and the fused distribution with a negative log-likelihood. The moderate-to-high head sees only moderate and high examples and adds a positive–negative margin to its binary cross-entropy, weighting moderate examples scored as high more heavily. $\mathcal{L}_{\text{type}}$ combines the hierarchy’s conditional losses, a flat classification loss, binary supervision for nontrivial combinations, and consistency penalties tying the combination branch to the type predictions. The novelty score regresses ν^* , the continuous target that $\text{raw}(x)$ in the novelty rule of Sec. 2.1 forms from the annotated coverage, substantive difference, evidence sufficiency and surface change of the target, and that the dataset thresholds into the four grades.

Scalar heads: level and ranking terms. Let ℓ_{δ} denote smooth- L_1 loss and w_x the example weight. For $a \in \mathcal{A}$, the level loss is $\mathcal{L}_a = \sum_x w_x \ell_{\delta}(\hat{a}_x - a_x^*) / \sum_x w_x$, where a_x^* is the annotation and $\hat{a}_x$ the prediction. The ranking term takes within-minibatch pairs separated by $|a_x^* - a_{x'}^*| \geq 0.05$ and contributes $[0.02 - \text{sgn}(a_x^* - a_{x'}^*)(\hat{a}_x - \hat{a}_{x'})]_+$, weighted by $\sqrt{w_x w_{x'}}$; it is dropped when a minibatch holds no eligible pair.

Ranking is not redundant with the level loss. At the transition δ and target spread used here the level loss operates in its L_1 regime, whose optimum is the median of the target, so a head supervised by it alone drifts towards a near-constant output—the evidence head most visibly. Ranking is scale-free and counteracts that drift, which is also why it takes most of the weight inside $\mathcal{L}_{\nu}$: thresholding reads the grade off the score, so only order matters there.

Coverage, decomposition, and variance terms. Beyond its level loss against the annotated joint value C^* , $\mathcal{L}_C$ adds a consistency loss against the upstream score C_J at relative weight 0.2,

reweighted by $1 - H_J$ so that examples with uncertain upstream coverage count for less; no unit-level support matrix is supplied as a target. $\mathcal{L}_{\text{dec}}$ combines three terms: attention regularization penalizes the squared entries of $\tilde{\alpha}\tilde{\alpha}^\top - I$ for the row-normalized attention matrix $\tilde{\alpha}$, plus half the mean off-diagonal cosine similarity; query diversity penalizes the mean squared cosine similarity between distinct queries; and a usage penalty $D_{\text{KL}}(\rho\|\mathbf{u})$, with $\rho_m = \sum_j B_{mj} / \sum_{r,j} B_{rj}$ and $\mathbf{u}$ uniform over the M units, keeps units from going unused. The variance floor carries its own coefficient,

$$\mathcal{L}_{\text{var}} = \mathbb{E}_x [[\nu - \text{Var}_m(c_{x,m})]_+ / \nu], \quad (16)$$

and acts on the unit coverages rather than on the token assignments, keeping the c_m within an example from coinciding.

Curriculum and feature perturbation. For a randomly sampled fraction of training examples, values derived from annotated joint coverage replace the joint-coverage features and the associated probability and entropy features—with probability 0.50 in epoch 1, 0.20 in epoch 2, and 0 thereafter. Development and test runs always use the upstream joint predictions. Selected joint-coverage features on the global-context path also receive Gaussian noise of standard deviation 0.03 and independent feature dropout at probability 0.10, which leaves untouched both the direct coverage inputs to the novelty heads and the entropy input to the fusion gate.

Inputs and initialization. The input adapter reconstructs pair probabilities missing from a ranking record from the available coverage label and score; pair labels and scores also appear in the pair text input. The two facet embeddings entering z_G have dimension 32 each. The model uses DeBERTa-v3-base with hidden dimension 768, maximum pair-input length 384, $K = 5$ and $M = 8$; the residual encoder is initialized from the learned joint model, and upstream checkpoints stay fixed. Padded priors have zero support, support values are capped at 0.995, and an empty evidence set falls back to a target-containing input with all prior support masked out. Training uses refinement weights $(\alpha_D, \alpha_O, \alpha_{MH}) = (0.40, 0.30, 0.30)$ and mixture weights $(\lambda_D, \lambda_O, \lambda_R) = (0.50, 0.35, 0.15)$; development-set calibration selects $(0.60, 0.20, 0.20)$ and $\theta_{MH} = 0.32$ for inference, fixed thereafter. The reported checkpoint was selected on development data at epoch 5 of 8.

D EXPERIMENTAL DETAILS

D.1 SETUP

Splits. The split is a date-ordered 13,815/3,944/3,352 partition of the 21,111 idea-level samples in IDEADELTA.

Configuration. All runs share one configuration and vary a single parameter at a time: deberta-v3-base backbone (He et al., 2023), 8 epochs, batch size 2, learning rate 1.5×10^{-5} with an encoder learning-rate ratio of 0.2, eight semantic units with slot competition enabled, and a fixed joint-prediction checkpoint supplying the encoder initialisation. The checkpoint reported in the main text is selected on dev; test is traversed once, at final evaluation, and is not used to select a checkpoint.

Dense retrieval baseline. The dense-RAG rows of Table 1 replace the stored candidate lists with retrieval performed at query time (Karpukhin et al., 2020). We embed every canonical idea with bge-large-en-v1.5 (Xiao et al., 2024), index the ideas whose date precedes the target’s, query the index with the target text, and pass the top five hits to the same single-call prompt the *direct* rows use. Nothing else changes: the same backbones, the same prompt, the same labels and the same scoring script. Each domain is indexed separately, so a target retrieves within its own domain as every other row does. Inside that index the date is the only constraint applied: the search ranges over all aspects, whereas the stored candidate lists come from aspect-restricted nearest-neighbour recall together with the cross-type pass of §2.

Metrics. We use four evaluation metrics, computed over the four ordered grades LOW, WEAK, MODERATE, HIGH, indexed 0 to 3. **Accuracy** is the fraction of ideas whose predicted grade equals the gold grade. It weights every example equally and does not distinguish errors by grade distance,

so frequent classes can dominate it. **Macro-F1** averages the per-class F_1 over the four grades with equal weight, which charges a system for abandoning a small class rather than letting the large ones absorb the loss. **QWK** is quadratic weighted kappa (Cohen, 1968),

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \quad w_{ij} = (i - j)^2 / 9,$$

where O is the confusion matrix and E the matrix expected from the two marginals if prediction and gold were independent. QWK measures chance-corrected ordinal agreement, penalizing distant grade errors more heavily than near misses. With a nonzero denominator, it equals 1 for exact agreement, 0 when observed weighted disagreement matches the chance reference, and is negative when disagreement exceeds that reference. A constant predictor has QWK 0 on our evaluation sets; independent random predictions need not yield exactly 0 in a finite sample. QWK depends on the assigned grades and their marginal distributions, so it does not isolate ranking quality from grade calibration. **MAE** (mean absolute error) measures the average absolute difference between predicted and gold novelty grades:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|,$$

where $\hat{y}_i$ and y_i are the predicted and gold grade indices in $\{0, 1, 2, 3\}$, respectively, and N is the number of evaluated ideas. Lower values indicate smaller grade errors. Among constant predictions, a median of the gold-label distribution minimizes MAE; this does not characterize the best input-dependent predictor. We therefore interpret MAE alongside the other metrics. Every table in the paper reports these four. On RINoBench the expert 1–5 scale is folded to the same four levels before any of them is computed, so every number in Table 2 is on the scale described here.

Per-domain metrics. Per-domain figures are recomputed from the stored predictions rather than read from the trainer, and reconcile with the trainer’s own pooled figures to within 5×10^{-5} on all four metrics.

D.2 RINoBENCH TRANSFER PROTOCOL

Labels and scale. The 277 test ideas of RINoBench (Schopf & Färber, 2026) carry expert novelty ratings on a 1–5 scale. We fold them to the four ordered grades above by merging 4 and 5, and every system in Table 2 — ours and the comparison rows alike — is scored on that folded scale by the same script that produces Table 1. The fold places 41.5% of the ideas in the top class.

Evidence and inference. Each target is paired with the 25 prior ideas the benchmark supplies, in place of the five of the IDEADELTA protocol. The checkpoint is the one reported in Table 1 and its trained input budget of $K=5$ is left unchanged, so the 25 priors are split into five blocks of five and the network runs once per block: every prior is read at the width the decomposition was trained for, and no input slot is repurposed. Each pass produces a continuous substantive-difference score for the target, the five scores are reduced to one, and three thresholds cut that score into the four ordered grades. No weight is retrained or fine-tuned on RINoBench data.

D.3 PARAMETER SENSITIVITY

Each sweep varies one parameter from the default configuration of Table 1 and holds the rest fixed: the upstream pair and joint products, the encoder initialisation, the evaluation files and the seed. Every point is a single run at seed 173, except for the seed sweep below.

Evidence budget: gains flatten beyond four input slots. Figure 3 varies the input-slot budget K over $K = 2, 3, 4, 5, 6, 8$. Accuracy rises steeply from $K=2$ to $K=4$ (0.731 $\rightarrow$ 0.806 on Graph ML, 0.716 $\rightarrow$ 0.772 on computer vision, 0.726 $\rightarrow$ 0.789 on time series) and then flattens across the three domains. The configured slot budget does not by itself establish how many non-padding priors a target receives. These curves show sensitivity to K under the evaluated setup; they do not distinguish limited evidence availability from limited benefit of additional evidence.

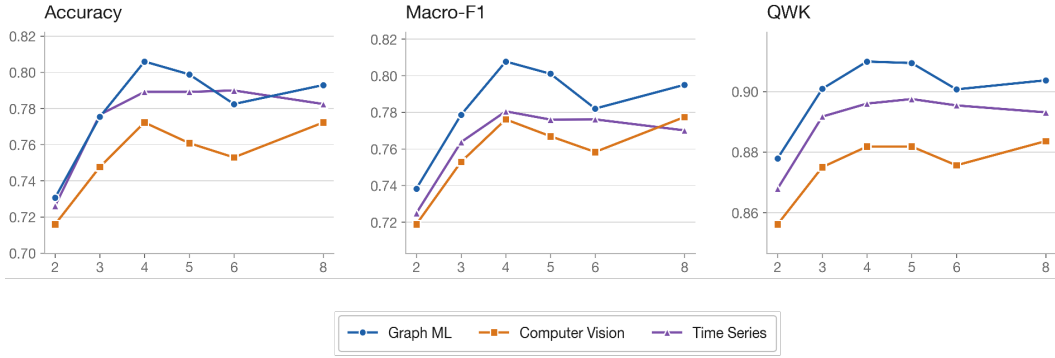


Figure 3: Sensitivity to the evidence budget K in the three domains. The default is $K=5$, the checkpoint of Table 1. K specifies the number of input slots, including padding when fewer priors are available.

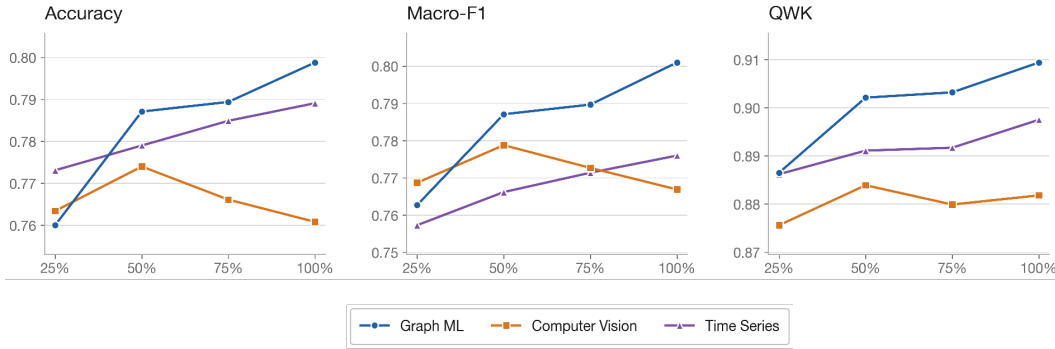


Figure 4: Sensitivity to the amount of EGRD training data in the three domains. Development and test sets stay fixed, and 100% is the default.

Training-set size: gains differ across domains. Figure 4 varies the number of training ideas over 25%, 50%, 75% and 100% of the training split, drawn as nested stratified prefixes so that consecutive points add samples without reordering what came before. Macro-F1 rises steadily with the fraction on Graph ML (0.7627 $\rightarrow$ 0.8010) and time series (0.7573 $\rightarrow$ 0.7760). Computer vision does not improve consistently beyond 50% and ends at 0.7669 against a peak of 0.7788, a gap of 0.012. More training data helps in most of the domains we measure, and the spread across the four fractions stays small: Accuracy moves by at most 0.039 in any domain.

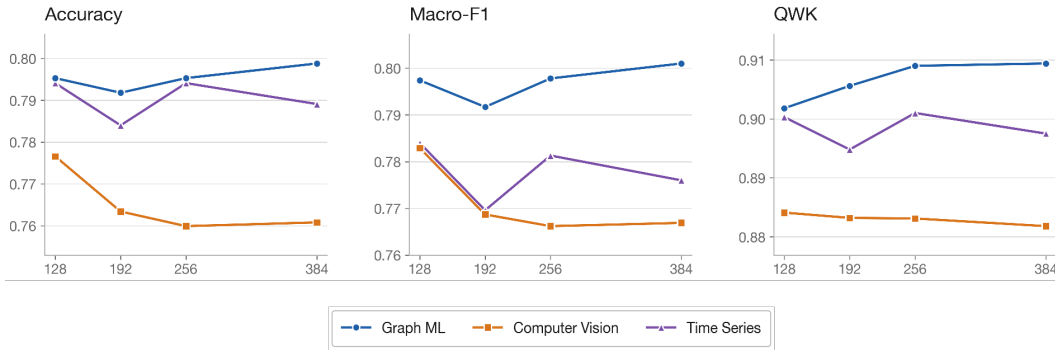


Figure 5: Sensitivity to encoder input length on the three domain test splits. The default is 384, and every setting scores the same test ideas and labels.

Encoder input length: limited sensitivity over the tested range. Figure 5 varies the encoder input length over 128, 192, 256 and 384 tokens. The four lengths separate the domains by little: no domain moves monotonically with input length, and tripling the input moves Macro-F1 by at most 0.017 in any domain. QWK spans 0.0076 on Graph ML, 0.0023 on computer vision and 0.0062 on time series across the four lengths, indicating that ordinal agreement changes little over the tested

input lengths. These results show limited performance sensitivity from 128 to 384 tokens; they do not establish how much of each input is retained at 128 tokens.

Table 6: Per-domain seed variability. Entries are sample standard deviations across seeds 173, 174 and 175 with identical data, code and configuration.

Domain	Accuracy	Macro-F1	QWK	MAE
Graph ML	0.0066	0.0050	0.0048	0.0067
Computer vision	0.0040	0.0021	0.0032	0.0027
Time series	0.0050	0.0045	0.0031	0.0051

Seed stability: the reported model is not a selected seed. Table 6 varies nothing but the random seed, running the default configuration on seeds 173, 174 and 175 with identical data and code and reporting the per-domain sample standard deviations. The spread is 0.004–0.007 in Accuracy, 0.002–0.005 in Macro-F1 and 0.003–0.007 in MAE, an order of magnitude below the ablation gaps of Table 3, so the headline numbers describe the configuration rather than one fortunate run, and the three sweeps above should be read by the shape of a curve rather than by the difference between adjacent points.

D.4 CASE-STUDY IDEAS

Construction of Figure 2. Panel (a) is read off the forward pass of the trained model and fits nothing. For the other two panels the test ideas are split 80/20 with the three cases held out, and every supervised step is fit on the 80% part alone. Panel (b) standardises z_R on that part, ranks its coordinates by univariate F -statistic against the grade, and shows the 140 that separate it best. Panel (c) reduces z_R to 50 principal components and projects it onto the first two axes of a shrinkage linear discriminant fit on the four grades. The probe reported in the panel is separate from both: a five-fold cross-validated multinomial logistic regression on standardised z_R , scored out of fold.

Texts. Table 7 gives the three targets of §4.4 and the five retained priors of each, with the date that makes the prior admissible, the coverage label the annotation carries, and the support A_k the model assigns. Ideas are stored as single contribution statements, so each row is the whole of the text the model reads for that idea.

E LIMITATIONS

Two bounds on the claims of this paper are worth stating directly.

Three domains. IDEADELTA is drawn from graph machine learning, computer vision and time series, and the RINoBench transfer (§4.2) indicates what lies outside them: roughly 30 of its 277 test ideas fall in these three fields, and the model reads the remainder without ever having been trained on their literature. Coverage of a field is what the decomposition needs, since a prior can only support a unit of a target the model is equipped to read, so we do not claim the reported Accuracy outside the three.

Transfer to expert judgment is partial. The transfer experiment of §4.2 shows the frozen checkpoint leading on the rank-sensitive metrics at 0.2245 QWK, but 0.2245 is a modest level of agreement, and Accuracy adds nothing to it: our row reaches exactly the constant predictor’s 0.4152 with a prediction that is not constant. What transfers is the ordering of ideas, not the grade an idea receives, and the label source, the domain mix and the folded scale all shift at once, so the result is partial agreement with expert judgment rather than evidence that the two scales coincide.

Table 7: The three case-study targets and their retained priors, verbatim. Coverage labels are *large*, *partial*, *related* and *none*; A_k is the support the model assigns to that prior.

Idea	Date	Cover	A_k	Text
<i>Graded low</i>				
CL.26954	2026-08	—	—	Propose an action-interleaved transformer architecture that explicitly models state transitions by interleaving state and action tokens in causal order to learn controllable dynamics.
CL.14929	2024-01	large	0.838	Introduce specialized action tokens to denote embodied interactions and state tokens to represent subsequent multisensory observations, enabling end-to-end instruction tuning for active exploration loops.
CL.17158	2025-02	partial	0.447	Employ a causal transformer with a query-based parallel representation updating mechanism to jointly model the dynamics of RGB-D data, 3D flow, and robotic actions in an end-to-end manner.
CL.13226	2022-03	partial	0.507	Design an action module that learns to predict state transitions in the latent space of the environment representation in an unsupervised manner to enable interactive control.
CL.8741	2023-06	related	0.070	Propose a generative pre-trained Transformer-based dynamics model to learn soft object behavior from large-scale interaction data.
CL.10777	2018-01	none	0.003	Propose a unified framework that integrates a temporal Conditional Random Field with discriminative sparse coding to jointly model action transitions and learn robust frame-level representations for fine-grained segmentation.
<i>Graded moderate</i>				
CL.18669	2026-01	—	—	Introduce token-level history caching to reuse previously generated image tokens directly, preventing identity drift caused by repeated image reconstruction in multi-round interactions.
CL.14692	2024-01	partial	0.523	Propose an adaptive token propagation mechanism that dynamically pauses and restarts tokens across layers to enable reuse of previously discarded information, addressing the inefficiency of permanent token removal.
CL.17874	2025-03	related	0.026	Propose a content-aware token aggregation mechanism that shares token centers across all image tokens and updates them exclusively during the training phase to eliminate inference overhead while capturing long-range dependencies.
CL.15897	2024-03	related	0.043	Propose a nested tokenization framework that hierarchically organizes image patches into region-level and global-level tokens to enable end-to-end processing of extremely large images without quadratic memory growth.
CL.12981	2022-02	related	0.071	Propose a token reorganization mechanism that dynamically preserves attentive image tokens and fuses inattentive ones into a single token during the feed-forward process to reduce computational redundancy.
CL.14768	2024-01	none	0.006	Introduce an iterative refinement strategy using weak noise levels to correct visual artifacts from 3D projection while preventing structural distortion and identity loss.
<i>Graded high</i>				
CL.18985	2026-02	—	—	Design a policy optimization algorithm that utilizes triplet-based trajectory mixing and contrastive response maximization to explicitly address high intra-class variance and low inter-class variance in fine-grained tasks.
CL.3103	2025-11	related	0.005	Introduce a reward aggregation strategy for deep reinforcement learning that averages returns over equivalent expression sequences to reduce training variance.
CL.18341	2026-01	related	0.007	Develop a unified reinforcement fine-tuning framework that integrates multi-reward normalization and length-constrained ensemble rewards to optimize fine-grained visual classification while suppressing excessive textual reasoning.
CL.17358	2025-02	none	0.002	Develop a reward-driven training strategy that uses student performance improvements and reduced teacher-student discrepancies as feedback signals to optimize the weight generation agent via policy gradient.
CL.14068	2023-02	none	0.002	Introduce a center-concentrated multi-prototype generation strategy to mitigate intra-class sparsity by minimizing intra-class distances and maximizing inter-class separation.
CL.12524	2021-03	none	0.003	Propose a quality estimation mechanism that calculates the distribution distance between intra-class and inter-class similarities to generate pseudo-labels for unsupervised training.